



Capstone Project 2

CMU-SE 451

Project Proposal

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Integrated School Safety Monitoring System

Submitted by C2SE.41

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PROJECT INFORMATION

Project acronym	PupilLinker		
Project Title	PupilLinker: Integrated School Safety Monitoring System		
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TABLE OF CONTENT

1. Project title	1
2. Team members.....	1
3. Supervisor(s).....	1
4. Problem statement	1
5. Survey / existing solutions.....	3
6. Objectives and scope	9
7. Key features & requirements	12
7.1. Functional requirements.....	12
7.2. Non-Functional requirements	15
8. Target users / stakeholders.....	17
9. Technology stack	18
10. Methodology & Development plan	19
10.1. Methodology	19
10.2. High-Level development plan.....	20
11. System architecture overview	22
11.1 System architecture description:	22
11.2 System context diagram	22
11.2.1. System context description.....	24
12. Potential risks and mitigation strategies	29
13. Project constraints.....	30
14. Expected outcomes / deliverables.....	32
15. References.....	33

TABLE OF FIGURES

Figure 1. System context diagram	24
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TABLE OF TABLES

Table 1. Team members	1
Table 2. Supervisor(s).....	1
Table 3. Existing solutions	8
Table 4. Potential risks and mitigation strategies	29
Table 5. Project constraints.....	30
Table 6. References	33

1. Project title

- PupilLinker: Integrated School Safety Monitoring System

2. Team members

Table 1. Team members

Name	Student ID	Role
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3. Supervisor(s)

Table 2. Supervisor(s)

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4. Problem statement

In the modern educational landscape, ensuring the safety and well being of pupils during school hours, transportation, and transitions between home and school has become one of the most pressing concerns for parents, educators, and school administrators alike. With increasingly busy lifestyles, the challenge of managing student pickup and drop off, monitoring attendance, communicating emergencies, and ensuring that children are always accounted for has grown significantly.

To better understand the seriousness and urgency of the problem, we can consider recent statistics on pupil safety, school incidents, and child related accidents in Vietnam and globally:

- Pupil safety incidents:

- + According to statistics from the ministry of public security, from december 2023 to december 2024, there were 4,111 traffic accidents involving school age children, accounting for 17.45% of all traffic accidents nationwide [1].
- + In the 2024-2025 school year, 365 cases of students violating traffic safety laws were handled, including not wearing helmets, underage driving, and running red lights [1].

- + More than 370,000 children suffer injuries in Vietnam every year. Official statistics indicate a decline of about 4% in accident caused child fatalities over recent years, translating to about 100 fewer child deaths annually [2].
- + Nearly 1,600 violent incidents, mainly fights, are reported to occur on school premises or outside school gates each year in Vietnam [3].
- Child drowning and safety:
 - + In Vietnam, about 2,000 children drown every year, making drowning the leading cause of death for children under the age of 19 [4], [5].
 - + In 2021, Vietnam recorded 4,019 drowning deaths, with 40% involving children under the age of 14 [6].
 - + Each year, approximately 700 children die from drowning, while only 33.75% of students know how to swim and only about 8.6% of schools have swimming pools [1].
- School home communication gaps:
 - + In Vietnam, school bus safety remains a top concern. In August 2019, a first grader in Hanoi died after being abandoned in a school bus for nine hours. In September 2020, another child was left alone on a school bus [7].
 - + In Hanoi alone, there are currently more than 2,000 school buses, and tens of thousands of students use public transport regularly [7].
 - + Globally, approximately 108 people are killed and 13,200 people are injured in school bus accidents each year (US statistics, 2013-2022) [8].
- Education scale and digital transformation:
 - + In the 2024-2025 school year, Vietnam has 25,255,251 pupils and students across 53,979 educational institutions [9].
 - + Primary school has the largest number with over 8.8 million students, followed by lower secondary with 6.5 million and preschool with 4.8 million children [10].
 - + The Vietnamese government has set the education development strategy through 2030, with an orientation toward 2045, aiming to elevate the education system to advanced Asian nation standards [10].

- Survey:

- + A significant gap exists in the Vietnamese market for a comprehensive pupil safety management solution. While vnEdu Connect provides basic attendance tracking and school parent communication, it lacks GPS based child tracking, real time bus monitoring, emergency alert systems, and secure pickup management features.
- + International solutions like Brightwheel, Procare, PikMyKid, Here Comes The Bus, and HopSkipDrive each address specific aspects of child safety management but are not available or adapted for the Vietnamese market, and none provides an all in one solution covering all aspects of pupil safety.

Current solutions in the market either focus on academic management with limited safety features (vnEdu Connect), operate exclusively in Western markets (Brightwheel, Procare, PikMyKid, Here Comes The Bus, HopSkipDrive), or lack the comprehensive integration needed for complete pupil safety management. This creates difficulties for Vietnamese parents, schools, and administrators in ensuring continuous child safety throughout the school day.

PupilLinker was created to thoroughly address these issues by providing a comprehensive pupil safety management platform, connecting parents, schools, teachers, bus drivers, and administrators through a unified system. By integrating real time GPS tracking, secure digital check in/check out, emergency alert systems, authorized pickup verification, school bus monitoring, and seamless parent school communication, PupilLinker will become the trusted companion ensuring every child's safety throughout their educational journey, bringing maximum peace of mind and convenience to parents and educators.

5. Survey / existing solutions

This section provides a comprehensive analysis of existing pupil safety and school management solutions in both the Vietnamese and international markets. By examining six key applications: vnEdu Connect (Vietnam), Brightwheel (USA), Procare (USA), PikMyKid (USA), Here Comes The Bus (USA), and HopSkipDrive (USA), we can better understand the competitive landscape and identify opportunities for differentiation.

- vnEdu Connect application: vnEdu Connect is a Vietnamese mobile application developed by VNPT Group as part of the vnEdu 4.0 education ecosystem. It serves as a connection platform between families and schools, currently deployed in thousands of schools nationwide serving millions of users.

Key features:

- + Attendance tracking and detailed timetables.
- + Grade lookup and academic performance monitoring.
- + School notifications and news updates.
- + Parent teacher messaging.
- + Tuition payment online.
- + Meal menu tracking.
- + Medication reminders and health monitoring.
- + Photo album sharing.
- + Authorized pickup registration (basic).

Limitations:

- + Overwhelmingly negative user feedback regarding login and access issues.
 - + Mandatory payments for basic features like viewing grades.
 - + Complex registration processes.
 - + Lacks GPS based real time child tracking.
 - + No school bus monitoring.
 - + Limited emergency alert capabilities.
 - + No secure biometric or photo based pickup verification.
 - + Not designed as a safety first platform.
- Brightwheel application: Brightwheel is the #1 rated childcare management software in the United States, trusted by millions of educators and families. It provides an all in one platform for childcare centers, preschools, and daycares.

Key features:

- + Digital check in/check out with QR code scanning.
- + Billing and automated payment processing.
- + Daily activity reports with photos and videos.
- + Parent teacher messaging.

- + Lesson planning and curriculum management.
- + Attendance tracking with room ratio monitoring.
- + Health screening.
- + Incident reporting.
- + Staff timecard management.
- + Assessment and developmental observations.

Limitations:

- + Primarily designed for childcare/daycare centers.
 - + Not K 12 schools.
 - + Lacks school bus GPS tracking.
 - + No emergency reunification features.
 - + Limited to English and Spanish language support.
 - + Subscription based pricing not suitable for Vietnamese market.
 - + No digital hall pass or campus movement tracking.
- Procure childcare application: Procure Solutions is a cloud and mobile-based childcare management software with over 30 years of experience, serving over 100,000 child care centers.

Key features:

- + Contactless check in/check out via QR codes or GPS curbside.
- + Tuition collection and billing management.
- + Parent engagement with real time updates.
- + Classroom management.
- + Staff to child ratio monitoring.
- + Incident recording and reporting.
- + Calendar and event management.
- + Family communication via messaging and email.
- + Integration with QuickBooks and payroll systems.

Limitations:

- + Similar to Brightwheel in focus (childcare/daycare).
- + Complex navigation and occasional glitches reported.
- + Lacks school bus tracking capabilities.

- + No emergency panic button or reunification tools.
- + Not available for Vietnamese market.
- + Limited transportation safety features.
- PikMyKid application: PikMyKid is the leading school safety and dismissal platform in the United States, used by thousands of schools, specifically designed for K 12 school environments.

Key features:

- + Dismissal management with real time car line coordination.
- + Parent app for instant dismissal plan changes.
- + Emergency notification and silent panic button.
- + Student reunification during emergencies.
- + Digital hall pass for campus movement monitoring.
- + Photo-based guardian identification verification.
- + Attendance tracking;
- + School messenger for push notifications.
- + After school program management.
- + Early release scheduling.

Limitations:

- + Focused primarily on US school systems.
- + No school bus GPS real time tracking.
- + Lacks health monitoring and meal tracking.
- + Limited academic progress tracking.
- + Not available in Vietnamese market.
- + No bilingual Vietnamese/English support.
- Here Comes The Bus application: Here Comes The Bus is an award winning school bus tracking app used by over 1,100 school districts across the US, providing parents real time bus location and status alerts.

Key features:

- + Real time GPS bus location tracking on map.
- + Push notifications when bus approaches stop.
- + Student ridership scanning (board/alight confirmation).

- + Customizable proximity alerts.
- + Multiple bus tracking for families with children on different routes.
- + Schedule change notifications.
- + HTTPS encrypted and secure communication.

Limitations:

- + Exclusively focused on bus tracking only.
 - + No attendance management, classroom features, or academic tracking.
 - + Requires school district purchase.
 - + Frequent issues with bus substitutions affecting accuracy.
 - + No emergency alert or dismissal management.
 - + Not designed for Vietnamese market.
- HopSkipDrive application: HopSkipDrive is a technology company providing safe, supplemental student transportation through a network of highly vetted CareDrivers, supporting over 10,000 schools across 17 US states.

Key features:

- + 15 point CareDriver certification and background checks.
- + Real time ride tracking with safe ride technology.
- + Two factor authentication for rider verification (name and code word).
- + Caregiver check in notifications.
- + In vehicle recording capability.
- + Proactive ride intervention by dedicated safe ride support team.
- + Wheelchair accessible vehicle options.
- + Trauma informed driver training.
- + Ride booking with 6 hour notice capability.

Limitations:

- + Operates as a ride sharing model, not a school management platform.
- + Very expensive for individual transportation.
- + Focused on supplemental transportation gaps.
- + No school wide attendance or academic management.
- + Limited to US markets.
- + Not a comprehensive pupil safety solution.

Existing solutions:

The PupilLinker application will be different and better than existing solutions in the following points:

Table 3. *Existing solutions*

Criteria	vnEdu Connect app	Brightwheel app	Procure app	PikMyKid app	Here Comes The Bus app	HopSkipDrive app	PupilLinker app
Target market	Vietnam - school-parent communication.	USA - childcare centers and daycares.	USA - childcare and daycare management.	USA - K 12 school safety and dismissal.	USA - school bus tracking.	USA - supplemental student transportation.	Vietnam - comprehensive K 12 pupil safety.
Safety focus	Basic attendance only.	Check-in/out with health screening.	Contactless check in, ratio monitoring.	Dismissal, emergency, hall pass, panic button.	Bus tracking and ridership scanning.	Ride safety with driver vetting.	Complete safety: GPS tracking, emergency alerts, secure pickup, bus monitoring, hall pass.
Real-time tracking				Limited (car line only)	Bus GPS tracking only	Full ride GPS tracking	Full GPS tracking for bus, children, and pickup.
Emergency features				Silent panic button, reunification.		Safe Ride Support team.	Emergency alerts, panic button, reunification, first responder notification.
Pickup verification	Basic pickup registration.	QR code check in/ check out.	QR/GPS curbside check in.	Photo ID verification of guardians.	Student scan on/off bus.	Code word + name verification.	Multi factor: QR code, code word, photo ID.
Communication	School notifications, messaging.	Messaging, daily reports, photos.	Messaging, email, calendar.	School messenger, push notifications.	Push/email notifications.	Caregiver notifications.	Real time messaging, push notifications, emergency broadcasts.
Language support	Vietnamese	English, Spanish	English	English	English	English	Vietnamese, English (bilingual)

PupilLinker competitive advantages:

- Based on our comprehensive analysis of existing solutions, PupilLinker offers several distinct competitive advantages:
 - + Comprehensive safety integration: Unlike any single existing solution, PupilLinker combines attendance management, real time GPS tracking, school bus monitoring, secure pickup verification, emergency alert systems, digital hall passes, and parent school communication into one unified platform.
 - + Vietnam market focus: PupilLinker is built specifically for Vietnam's educational environment, supporting Vietnamese/English bilingual interface and adapted for the Vietnamese school system structure.
 - + Advanced safety technology: Multi factor pickup verification combining QR codes, code words, and photo ID verification provides the most secure child handover process available.
 - + Real time everything: Real time GPS bus tracking, real time student location during school hours, real time attendance updates, and real time emergency broadcasting ensure parents and schools always know where children are.

6. Objectives and scope**General objectives**

Develop a specialized, reliable, and easy to use PupilLinker mobile application, providing comprehensive pupil safety management services for kindergartens, primary schools, and secondary schools across Vietnam. The goal is to become the leading pupil safety application, trusted by parents and educators to ensure every child's safety throughout the school day.

Specific objectives

- Phase 1 (MVP - Minimum viable product) - Sprint 1 & Sprint 2 (Weeks 2-7):
 - + Develop a cross-platform mobile application: Build the PupilLinker application on both Android and iOS with an intuitive and easy to navigate interface optimized for parents, teachers, and administrators.
 - + User management system: Implement features for registration, login, personal profile management for multiple user roles (parent, teacher, school administrator, bus driver), and secure password recovery.

- + Digital attendance system: Allow teachers to record student attendance via QR code scanning or manual entry. The system automatically notifies parents of their child's attendance status in real time.
- + Secure check in/check out system: Implement multi factor verification for student pickup including QR code scanning, photo ID verification of authorized guardians, and code word confirmation.
- + School bus GPS tracking: Enable real time GPS tracking of school buses, allowing parents to monitor bus location, receive proximity alerts, and confirm student boarding/alighting.
- + Emergency alert system: Develop a comprehensive emergency notification system with panic button for teachers, real time alerts to parents and administrators, and integration with emergency protocols.
- + Parent school communication: Build a messaging system for direct communication between parents and teachers, school wide announcements, and push notification capabilities.
- + PostgreSQL database: Build a stable and secure PostgreSQL database with PostGIS extension to store user information, student data, GPS tracking records, attendance history, and communication logs.
- Phase 2 (Expansion) - Sprint 3 & Sprint 4 (Weeks 8-13):
 - + Health monitoring module: Track student health records, medication reminders, allergy alerts, temperature screening logs, and illness tracking with parent notification.
 - + Meal and nutrition management: Display daily meal menus, track student meal consumption, support dietary restriction alerts, and provide nutrition information to parents.
 - + Incident reporting system: Enable teachers and staff to document and report incidents, injuries, and behavioral observations with photo evidence, parent notification, and administrative follow up tracking.
 - + Smart notification system: Scheduled reminders for events and activities, maintenance of notification preferences, and quiet hours management.

- + Activity and event management: Provide tools for schools to create, manage, and communicate events, field trips, extracurricular activities, and parent teacher conferences.
- + Analytics dashboard: Provide school administrators with comprehensive analytics on attendance patterns, safety metrics, communication engagement, and operational insights.

Scope**- In scope (Project scope):**

- + Development of the PupilLinker mobile application for Android and iOS operating systems.
- + Multi role user management system (parent, teacher, administrator, bus driver).
- + Digital attendance tracking and real time notification system.
- + Secure student check in/check out with multi factor verification.
- + Real time school bus GPS tracking and monitoring system.
- + Emergency alert and notification broadcast system.
- + Parent, school, teacher communication messaging system.
- + Student health monitoring and medication reminder module.
- + Meal menu and nutrition management.
- + Incident reporting and documentation system.
- + Activity and event management tools.
- + Analytics dashboard for school administrators.
- + PostgreSQL database with PostGIS for geospatial data.
- + Push notification system with priority management.
- + Multilingual user interface (Vietnamese and English).

- Out of scope (Outside project scope):

- + Development of applications for web or desktop platforms.
- + Integration with smart wearable devices (smartwatches, GPS trackers).
- + Online learning management system (LMS) features.

- + Integration with complex payment systems requiring advanced identity verification (KYC).
- + Advanced AI based facial recognition training models.
- + Integration with government education databases.
- + Social media features such as forums, post sharing, etc.

7. Key features and requirements

7.1. Functional requirements

- FR01: Multi role user account management
 - + Registration and login: Parents, teachers, administrators, and bus drivers can register using email/phone number or through Google OAuth. The system supports secure login with JWT token management and session control.
 - + Role based profile management: Each user role has a tailored profile with relevant fields. Parents manage child profiles and authorized pickup delegates. Teachers manage class rosters. Administrators oversee school wide settings. Bus drivers manage route information.
 - + Password recovery: Secure password recovery mechanism via email or SMS verification code.
 - + Pupil profile management: Parents and administrators can create and manage student profiles including personal information, medical conditions, allergies, emergency contacts, authorized pickup persons with photos, and transportation preferences.
- FR02: Digital attendance tracking system
 - + Attendance recording: Teachers can record attendance via QR code scanning, manual roll call, or NFC tap. Attendance is recorded with timestamp and location data.
 - + Automatic parent notification: Parents receive immediate push notifications when their child is marked present, absent, or late.
 - + Attendance analytics: The system provides attendance reports, trends, and alerts for chronic absenteeism patterns.

- FR03: Secure check in/check out system
 - + Multi factor pickup verification: The system verifies authorized guardians through a combination of QR code scanning, photo ID matching, and personal code word confirmation before releasing a student.
 - + Authorized person management: Parents can add, modify, or remove authorized pickup persons through the app, including scheduling temporary delegates and carpool arrangements.
 - + Dismissal management: Teachers and administrators can manage dismissal queues, track pickup status in real time, and coordinate car line flow efficiently.
 - + Early release management: Parents can schedule early releases through the app, requiring administrator approval and maintaining a complete audit trail.
- FR04: School bus GPS tracking and monitoring
 - + Real time bus tracking: Parents can view real time GPS location of their child's school bus on an interactive map with customizable proximity alerts.
 - + Student boarding confirmation: The system confirms when students board and alight from the bus, sending notifications to parents.
 - + Route management: Administrators can define, modify, and monitor bus routes with estimated arrival times at each stop.
 - + Bus driver app: Bus drivers have a dedicated interface for route navigation, student roster management, and boarding confirmation.
- FR05: Emergency alert system
 - + Panic button: Teachers and staff have access to a silent panic button that immediately alerts administrators and triggers emergency protocols.
 - + Emergency broadcast: Administrators can send real time emergency notifications to all parents, teachers, and staff with customizable message templates for different emergency scenarios.
 - + Emergency reunification: During emergencies, the system provides tools to account for every student, verify guardian identification, and track reunification progress.
 - + Location sharing during emergencies: The system shares relevant location data with emergency contacts and first responders when activated.

- FR06: Parent school communication system
 - + Direct messaging: Parents and teachers can communicate through secure in app messaging with text, photos, and document attachment support.
 - + School announcements: Administrators can send school wide or targeted announcements to specific classes, grades, or groups.
 - + Push notifications: The system sends push notifications for all important events including attendance updates, pickup confirmations, emergency alerts, and school announcements.
 - + Event calendar: Shared calendar for school events, parent teacher conferences, field trips, and activities with RSVP functionality.
- FR07: Digital hall pass system
 - + Hall pass issuance: Teachers can issue digital hall passes to students with destination, time limit, and purpose specified.
 - + Real time monitoring: Administrators can view all active hall passes campus wide, identifying students who are outside classrooms.
 - + Time tracking and alerts: The system automatically tracks hall pass duration and alerts teachers and administrators when passes expire.
- FR08: Health monitoring module
 - + Health profile: Parents can maintain student health profiles including medical conditions, allergies, medications, and emergency medical contacts.
 - + Medication reminders: The system supports medication schedule tracking with reminders for school nurses and notifications to parents.
 - + Temperature and health screening: Schools can log daily health screenings with automatic alerts for abnormal readings.
- FR09: Incident reporting system
 - + Incident documentation: Teachers and staff can document incidents with text descriptions, photo evidence, timestamp, location, and involved parties.
 - + Parent notification: Parents are automatically notified of incidents involving their child, with the ability to acknowledge and respond.
 - + Follow up tracking: Administrators can track incident resolution progress and maintain a complete incident history.

- FR10: Activity and event management
 - + Event creation and management: Schools can create events with details, location, time, and participant requirements.
 - + Field trip management: Manage field trip permissions, attendance tracking during trips, and GPS tracking of field trip buses.
 - + RSVP and sign up: Parents can RSVP for events, sign up for volunteer activities, and receive reminders.
- FR11: Meal and nutrition management
 - + Daily menu display: Schools can publish daily meal menus with nutritional information.
 - + Dietary restriction alerts: The system flags meals that conflict with student allergies or dietary restrictions.
 - + Meal tracking: Track student meal participation for billing and nutritional reporting.
- FR12: Multi language support
 - + The application must support at least two languages: Vietnamese and English, allowing users to easily switch between languages within the app.

7.2. Non Functional requirements

- NFR01: Performance
 - + Response time: The application must have a response time of under 3 seconds for critical safety actions such as emergency alerts, attendance recording, and pickup verification. General actions must respond within 5 seconds.
 - + GPS tracking refresh: Real time bus tracking must update every 10 seconds maximum.
 - + Performance measurement methodology: Response time will be measured using automated load testing tools (Apache JMeter, k6) under realistic conditions. Metrics include: P50, P95, P99 (Percentiles) latency percentiles measured at the API gateway level.
 - + Load capacity: The system must be able to handle at least 10.000 concurrent users without significant performance degradation. Load capacity measurement: Concurrent user capacity will be validated through stress testing with gradual

user ramp-up (100, 1.000, 5.000, 10.000 users). Success criteria: API error rate <1%, response time <3s for 95% of requests, server CPU utilization <80%, memory usage stable without memory leaks.

- + Resource optimization: The application must be optimized for efficient use of device resources (battery, mobile data), especially during continuous GPS tracking.

- NFR02: Usability

- + Intuitive interface: The user interface must be designed to be intuitive, clear, and easy to navigate for all user roles, especially for parents who may have varying levels of technical proficiency.
- + Simple processes: Critical safety actions (sending emergency alerts, recording attendance) must be completed in a maximum of 3 steps.
- + First time user experience: New users must be able to complete core tasks without external guidance within their first session.

- NFR03: Reliability

- + Uptime: The system must ensure a minimum uptime of 93% for safety critical features (emergency alerts, GPS tracking).
- + Data recovery: All data must be backed up hourly with full restoration capability.
- + Offline capability: Core safety features (emergency contacts, student information, QR code display) must function in offline mode.

- NFR04: Security

- + Data encryption: All sensitive data must be encrypted during transmission (TLS 1.3) and storage (AES-256).
- + Authentication: Multi factor authentication for administrator accounts. JWT-based session management with token refresh mechanism.
- + Location data protection: GPS data encrypted at rest and in transit, with strict access controls and 90 day retention limit.
- + Vietnam cybersecurity law compliance: Adhere to Decree 13/2023/ND-CP on personal data protection.

- NFR05: Scalability
 - + The system must be designed to support millions of users and thousands of schools nationwide in the future.
 - + PostgreSQL database with PostGIS must support horizontal scaling for geographic data.
- NFR06: Maintainability
 - + Source code must be well structured, adhere to coding standards, and have complete documentation.
 - + Automated test suites (unit tests, integration tests, end-to-end tests) must be maintained.
- NFR07: Compatibility
 - + The application must run smoothly on Android 8.0+ and iOS 14.0+.
 - + Compatible with various screen sizes and resolutions of different mobile devices.

8. Target users / stakeholders

- Target users:

- + Parents and guardians: Primary users who need to monitor their children's safety, receive real time updates on attendance, track school bus location, manage authorized pickup persons, and communicate with teachers and school staff.
- + Teachers and educators: Users who manage classroom attendance, issue digital hall passes, communicate with parents, document incidents, initiate emergency alerts, and manage dismissal processes.
- + School administrators: Users who oversee school wide safety operations, manage user accounts, configure emergency protocols, analyze safety analytics, and coordinate with all stakeholders.
- + School bus drivers: Users who navigate assigned routes, confirm student boarding and alighting, communicate with school administrators, and maintain real time location sharing.
- + School nurses and health staff: Users who manage student health records, medication schedules, and health screening logs.

- Stakeholders:

- + End users: Parents, teachers, administrators, and bus drivers who directly use the PupilLinker application.
- + Educational institutions: Kindergartens, primary schools, and secondary schools that adopt PupilLinker for pupil safety management.
- + Development team: Software engineers, UI/UX designers, testers, and the project manager responsible for the design, development, and maintenance of the PupilLinker application.
- + Guiding lecturers/advisors: Provide expert knowledge and guide the technical direction and project management throughout the development process.
- + Transportation service providers: School bus companies and drivers that provide student transportation services.
- + Potential partners: Educational technology companies, school insurance providers, and government education agencies.

9. Technology stack

- Frontend (mobile): Flutter (Dart).
- Frontend (Website): React.
- Database management system (DBMS): PostgreSQL with PostGIS extension.
- API and backend: Node.js with Express.js (RESTful APIs).
- Real time communication: Socket.IO for WebSocket connections.
- Message queue: Apache Kafka for event driven architecture.
- Caching: Redis for real time GPS data and session management.
- Cloud platform: Google Cloud Platform for deploying backend and database.
- Maps and location services: Google Maps SDK, Google Maps Directions API.
- Push notification service: Firebase Cloud Messaging (FCM).
- Authentication: JWT (JSON Web Tokens) with OAuth2 (Google).
- Development tools (IDE): Android Studio, Xcode, Visual Studio Code.
- Version control tool: Git, GitHub.
- Project management tool: Trello.
- UI/UX design tool: Figma.
- Diagramming tool: PlantUML Web Server, Draw.io.

- API testing tool: Postman.
- CI/CD: GitHub Actions

10. Methodology & Development plan

10.1. Methodology

The PupilLinker project will apply the Agile (Scrum) software development methodology with short iterative cycles (sprints). This method allows the development team to flexibly adapt to requirement changes, continuously receive user feedback, and improve the product through each phase. The main principles of Agile (Scrum) that will be followed include:

- + Iterative and incremental development: The project will be divided into small sprints, with each sprint creating a functional piece of software, gradually adding features. This helps minimize risk, allows for early feedback, and continuous product improvement.
- + Continuous collaboration and effective communication: Encourages daily interaction through scrum meetings (daily scrum), sprint planning, sprint review, and sprint retrospective. This ensures all team members are aware of the progress, issues that arise, and work together to find solutions.
- + Adaptability to change: Instead of following a rigid plan, the team will be ready to adjust requirements and priorities based on user feedback and market changes. The product backlog will be continuously updated to reflect the latest requirements.
- + Frequent delivery: Functional software versions will be delivered periodically (at the end of each sprint) so that stakeholders can review and provide timely feedback. This helps ensure the final product meets the users' actual needs.
- + Self organizing and cross functional teams: The development team will decide the best way to complete the work, and team members will have sufficient skills to perform various tasks throughout the product development lifecycle.
- + Transparency: All aspects of the development process (progress, issues, solutions) are made public and transparent to all stakeholders.

10.2. High-Level development plan**- Week 1 (January 27 - February 2, 2026) Inception & Planning:**

- + Complete initial project documents including proposal, project plan, and product backlog.
- + Define detailed requirements and user stories for all features.
- + Design high level system architecture and database schema.
- + Set up development environment, CI/CD pipeline, and project management tools.
- + Create initial database schema with PostgreSQL and PostGIS.
- + Set up Git repository structure and branching strategy.

- Weeks 2-4 (Sprint 1, February 3 - February 23, 2026):

- + Detailed UI/UX design for core screens: registration, login, profiles, dashboard.
- + Backend development: Authentication, multi role user management APIs, JWT implementation.
- + Frontend development: User registration, login, profile management interfaces.
- + Backend development: Digital attendance tracking APIs with QR code generation.
- + Frontend development: Teacher attendance interface, parent attendance notification view.
- + Unit, integration, and security testing for Sprint 1 features.

- Weeks 5-7 (Sprint 2, February 24 - March 16, 2026):

- + Backend development: Secure check in/check out system with multi factor verification APIs.
- + Frontend development: Pickup verification interface, authorized person management.
- + Backend development: School bus GPS tracking APIs, real time WebSocket connections.
- + Frontend development: Map based bus tracking interface, proximity alert configuration.
- + Backend development: Emergency alert system APIs, panic button mechanism.
- + Frontend development: Emergency alert interfaces for all user roles.
- + Integrate push notification system via Firebase Cloud Messaging.
- + Functional and performance testing for Sprint 2 features.

- Weeks 8-10 (Sprint 3, March 17 - April 6, 2026):

- + Backend development: Parent school communication system, messaging APIs.
- + Frontend development: Chat interfaces, announcement boards, event calendar.
- + Backend development: Digital hall pass system APIs with real time monitoring.
- + Frontend development: Hall pass issuance and monitoring interfaces.
- + Backend development: Health monitoring module, incident reporting APIs.
- + Frontend development: Health profile, medication reminders, incident reporting forms.
- + Testing and optimization for all Sprint 3 features.

- Weeks 11-13 (Sprint 4, April 7 - April 27, 2026):

- + Backend development: Meal and nutrition management, activity/event management APIs.
- + Frontend development: Meal menu display, event management interfaces.
- + Backend development: Analytics dashboard APIs with data aggregation.
- + Frontend development: Administrator analytics dashboard.
- + UI/UX refinement based on feedback from previous sprints.
- + Performance optimization: database queries, caching strategies, resource usage.
- + Comprehensive testing: unit, integration, system, and UAT testing.

- Week 14 (April 28 - May 4, 2026) Testing, debugging, and final adjustments:

- + Comprehensive end to end testing across all features.
- + Security penetration testing and vulnerability assessment.
- + Bug fixing and stability improvements.
- + Compatibility testing across multiple devices and OS versions.
- + Performance load testing with simulated concurrent users.

- Week 15 (May 5 - May 11, 2026) Final deployment and presentation:

- + Prepare and finalize all technical documentation.
- + Complete all capstone 2 materials.
- + Prepare user guide documentation.
- + Prepare progress and final reports.
- + Prepare final presentation slides and demo.

11. System architecture overview

11.1 System architecture description:

- PupilLinker application will be designed based on microservices architecture. The decision to adopt microservices architecture was based on several critical factors:
 - + Independent scalability: Pupil safety services experience variable traffic patterns. During school hours, the GPS tracking and attendance services handle significantly higher loads than the analytics service. Microservices allow each service to scale independently based on demand.
 - + Fault isolation and high availability: For a pupil safety application, system availability is critical. If the meal management service encounters an issue, it should not affect the core emergency alert or GPS tracking functionality. Microservices provide fault isolation.
 - + Technology flexibility: Different services have different requirements. The location service benefits from geospatial optimizations (PostGIS), the real time tracking uses Redis caching, and the notification service integrates with Firebase FCM.
 - + Independent deployment: The team can develop, test, and deploy individual services independently, enabling CI/CD practices for faster feature releases.
 - + Real time requirements: The pupil safety application requires real time GPS tracking with <3 second response times. Microservices architecture with Redis caching and dedicated Location/Tracking services enables this performance.
- Core microservices include:
 - + User service: Authentication, authorization, multi role profile management.
 - + Attendance service: Digital attendance recording, QR code management, attendance analytics.
 - + Safety service: Check in/check out verification, pickup management, guardian authentication.
 - + Location service: GPS tracking for buses and students, geofencing, PostGIS spatial queries.
 - + Emergency service: Panic button handling, emergency broadcasting, reunification management.

- + Communication service: Messaging, announcements, push notifications via Firebase FCM.
- + Health service: Health profiles, medication reminders, screening logs, incident reports.
- + Event service: Activity management, field trips, calendar events, RSVP management.
- + Nutrition service: Meal menu management, dietary tracking, allergy alerts.
- + Analytics service: Data aggregation, reporting, dashboard metrics.
- + API gateway: Request routing, rate limiting, authentication, load balancing.

11.2 System context diagram

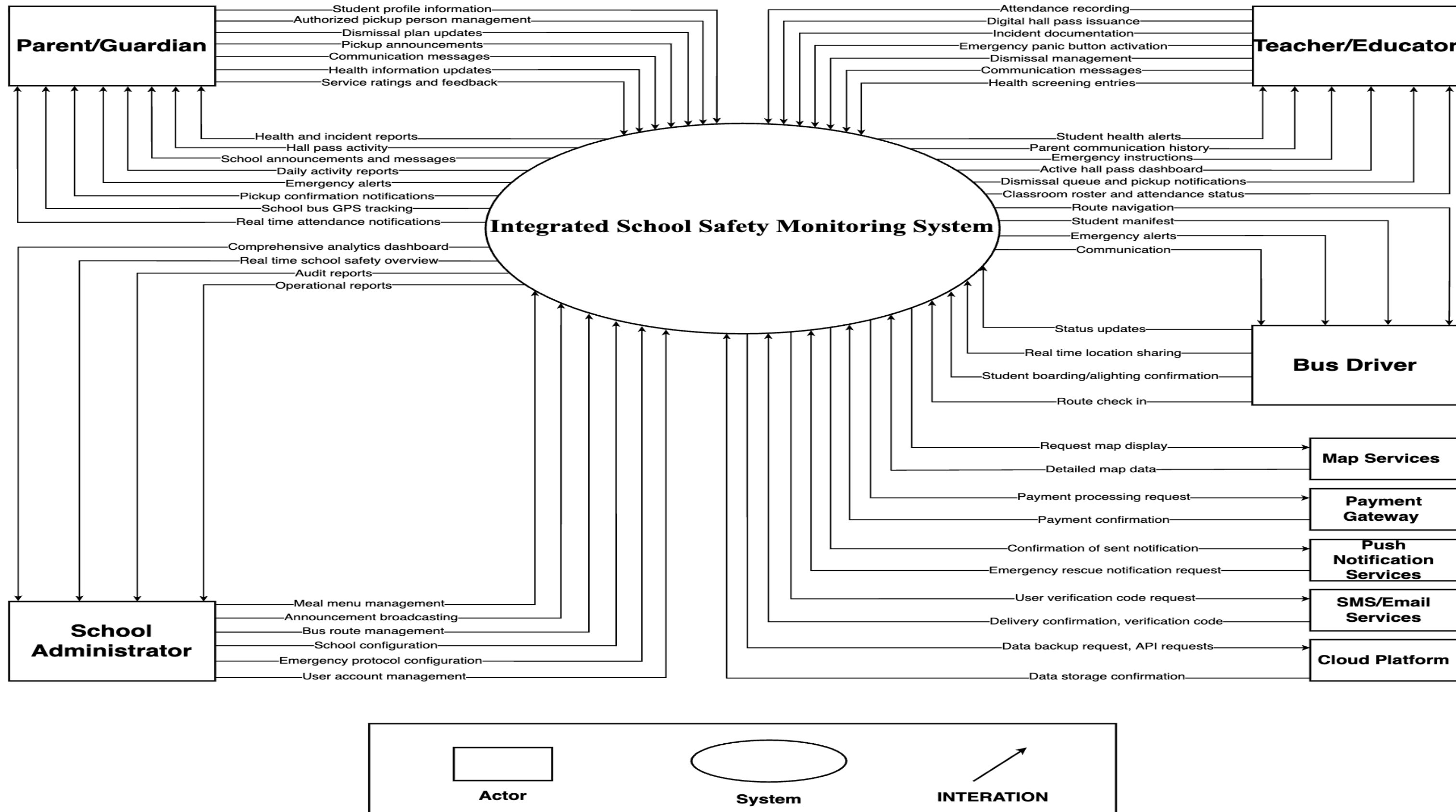


Figure 1. System context diagram

11.2.1. System context description**Parent/Guardian:****- Inputs to system:**

- + Registration/authentication data: Email, phone number, social media credentials for account creation..
- + Student profile information: Child's name, grade, class, medical conditions, allergies, emergency contacts, and registered vehicles.
- + Authorized pickup person management: Addition, modification, and removal of authorized guardians with photo IDs, relationships, and code words.
- + Dismissal plan updates: Daily transportation mode changes (car pickup, bus, walking, carpool), delegate assignments, early release requests.
- + Pickup announcements: Real time arrival announcements at school car line via app.
- + Communication messages: Direct messages to teachers, responses to announcements, RSVP for events.
- + Health information updates: Medication schedules, allergy updates, illness notifications, absence requests.
- + Service ratings and feedback: Ratings for bus service quality and app experience.

- Outputs from system:

- + Real time attendance notifications: Instant alerts when child is marked present, absent, or late.
- + School bus GPS tracking: Live map showing bus location, route, estimated arrival time at stops.
- + Pickup confirmation notifications: Alerts confirming child has been picked up by verified guardian.
- + Emergency alerts: Immediate notifications during school emergencies with instructions and reunification information.
- + Daily activity reports: Summary of child's attendance, meals, health screenings, and incidents.
- + School announcements and messages: Push notifications for school news, events, and direct teacher communications.
- + Hall pass activity: Notifications when child leaves and returns to classroom.

- + Health and incident reports: Detailed reports of health screenings, medication administration, and any incidents.

Teacher/Educator:**- Inputs to system:**

- + Attendance recording: QR code scan, manual roll call for student attendance.
- + Digital hall pass issuance: Destination, time limit, purpose, and student selection for hall passes.
- + Incident documentation: Text descriptions, photo evidence, timestamps, involved parties for incident reports.
- + Emergency panic button activation: Silent emergency alert trigger with optional description.
- + Dismissal management: Student release confirmations, pickup queue management.
- + Communication messages: Direct messages to parents, class wide announcements, daily reports.
- + Health screening entries: Temperature readings, symptom observations, medication administration logs.

- Outputs from system:

- + Classroom roster and attendance status: Real time view of all student statuses.
- + Dismissal queue and pickup notifications: Real time car line status and authorized guardian arrivals.
- + Active hall pass dashboard: Live view of all students outside classrooms.
- + Emergency instructions: Step by step protocol guidance during emergencies.
- + Parent communication history: Message threads and announcement confirmations.
- + Student health alerts: Allergy alerts, medication reminders, and health condition flags.

School Administrator:**- Inputs to system:**

- + User account management: User verification, account creation, role assignments, suspensions.
- + Emergency protocol configuration: Emergency types, response templates, notification chains, reunification procedures.

- + School configuration: Class schedules, bus routes, school calendar, notification policies.
- + Bus route management: Route definitions, stop assignments, driver assignments, schedule updates.
- + Announcement broadcasting: School wide and targeted communications.
- + Meal menu management: Daily meal plans, nutritional information, dietary options.
- Outputs from system:
 - + Comprehensive analytics dashboard: Attendance rates, safety metrics, communication engagement, incident trends.
 - + Real time school safety overview: Campus wide student location status, active emergencies, bus fleet status.
 - + Audit reports: Detailed logs of all safety critical actions, pickup records, emergency responses.
 - + Operational reports: Staff performance, bus route efficiency, system usage statistics.

Bus Driver:

- Inputs to system:
 - + Route check in: Start and end route confirmations.
 - + Student boarding/alighting confirmation: Scanning or manual confirmation of students getting on/off bus.
 - + Real time location sharing: Continuous GPS coordinates transmitted to the system.
 - + Status updates: Route delays, vehicle issues, schedule changes.
- Outputs from system:
 - + Route navigation: Optimized route display with stops and schedule.
 - + Student manifest: List of assigned students with boarding status at each stop.
 - + Emergency alerts: Immediate notifications of school emergencies affecting routes.
 - + Communication: Messages from administrators regarding route changes or updates.

External services

- Map services (Google maps API):
 - + Inputs to system: GPS coordinates, detailed map data, optimized route information, real time traffic updates, location based services.

- + Outputs from system: Location requests, route optimization queries, map display requirements.
- Payment gateway:
 - + Inputs to system: Payment confirmations, transaction status updates, payment receipts, security validations.
 - + Outputs from system: Payment processing requests, transaction data, user payment information.
- Push notification services (Firebase cloud messaging):
 - + Inputs to system: Confirmation of sent notification.
 - + Outputs from system: Push notification delivery for all user types across Android and iOS notifications
- SMS/Email services:
 - + Inputs to system: Delivery confirmation, verification code, message delivery status, communication log.
 - + Outputs from system: User verification code request, password reset link request, service confirmation email.
- Cloud platform:
 - + Inputs to system: Data storage confirmation, scalable resources, infrastructure services, security services, system availability assurance
 - + Outputs from system: Data backup requests, API requests, database operations

12. Potential risks and mitigation strategies

Table 4. Potential risks and mitigation strategies

Category	Risk	Probability	Impact	Exposure	Mitigation strategy
Technical	Difficulty in integrating real time GPS tracking with acceptable battery consumption across different devices.	3	4	12	Implement adaptive GPS polling (reduce frequency when bus is stopped). Use Google Maps SDK optimized settings. Test across multiple device models.
	Performance issues when handling concurrent GPS updates from multiple buses and thousands of parent requests.	3	4	12	Use Redis caching for real time location data. Implement WebSocket for efficient real-time updates. Load testing with realistic concurrent scenarios.
	Security vulnerabilities in multi factor pickup verification system.	2	5	10	Implement comprehensive security testing. Use industry standard encryption. Regular penetration testing. Follow OWASP security guidelines.
	Ensuring cross platform compatibility of QR code scanning.	3	3	9	Use widely supported Flutter plugins. Extensive device testing. Graceful degradation for unsupported features.
Schedule	Delay in completing features according to the 15 week plan. During project implementation, our team needs to learn and has more work to do.	4	4	16	Apply agile (scrum) methodology with short sprints. Prioritize core safety features for MVP. Maintain contingency plan.
	Lack of team experience with real time tracking and geospatial technologies.	4	3	12	Allocate time for learning PostGIS and WebSocket. Seek mentor support. Organize internal knowledge-sharing sessions.
Resource	Lack of resources (time, tools, budget) to complete the project.	3	4	12	Use open source tools and free tier cloud services. Detailed resource planning. Maximum use of available libraries.
	Difficulty in obtaining realistic test data for school environments (bus routes, student records, attendance patterns).	4	4	16	Create comprehensive mock data generators. Partner with local schools for anonymized test scenarios. Use synthetic data generation tools.
Data	GPS tracking accuracy issues in urban	3	3	9	Use google maps SDK with high accuracy mode.

	environments with tall buildings and tunnels.				Implement location smoothing algorithms. User adjustable accuracy settings.
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13. Project constraints

Table 5. Project constraints

Constraint	Constraints description	Guidelines for acceptance
Economic	The PupilLinker project is developed as an academic capstone project at Duy Tan University with limited financial resources. Development costs are minimized by utilizing open-source technologies (Flutter, Node.js, PostgreSQL) and free tier cloud services (Google Cloud Platform). The team of 5 members works without salary during the 15 week development period. Production costs are reduced through cross-platform development with Flutter. Operating costs focus on cloud hosting, API services (Google Maps), and database maintenance.	Elements for consideration are design costs, production costs, maintenance costs, operating costs, and sales price
Environmental	The PupilLinker application is designed to operate reliably across diverse environmental conditions in Vietnam. The mobile app must function in varying temperatures (tropical climate with high humidity), areas with unstable network connectivity, and under different signal conditions (4G/5G/WiFi). GPS accuracy considerations include urban environments with tall buildings. The application supports energy efficient operations to minimize battery consumption during continuous GPS tracking.	Impact of the design on the environment as well as impact of the environment (e.g. temperature range, humidity, vibration, electromagnetic interference immunity, and shock) on the design should be considered. Design for recycling and design to use recycled materials should also be considered
Ethical	PupilLinker adheres to strict ethical standards regarding child data privacy and security. All personal information is encrypted using HTTPS/SSL during transmission and secure database encryption for storage. The application complies with Vietnam Cybersecurity Law (Decree 13/2023/ND-CP) and implements transparent data collection policies. The platform ensures fair treatment of all users through transparent operations and clear terms of service. Child location data is treated with the highest level of sensitivity.	Ethical considerations can be broad. Areas that are typically addressed include intellectual property, reverse engineering, privacy, security, and the conflict between cost and safety
Public health, safety, and welfare	PupilLinker prioritizes student safety as its core design principle. The emergency alert feature is designed for instant access (maximum 2 steps). Real time GPS tracking	Includes safety standards as well as impact of the design on users (for example, electrical or physical hazards)

	allows parents to monitor their children's transportation. The platform verifies authorized pickup persons through multi factor authentication. Safety features include emergency notifications, guardian photo verification, and a comprehensive audit trail of all child related activities.	
Social and global	PupilLinker addresses significant safety needs in Vietnam where 4,111 traffic accidents involving school age children occurred in a single year. The application improves access to school safety technology for families across all economic levels. Multilingual support (Vietnamese and English) ensures accessibility for both local users and international families. The intuitive interface accommodates users of varying technical abilities.	Addresses aspects such as benefits, risks, the man machine interface, the acceptance of products by the intended user or by society at large, global and socially responsible engineering.
Cultural	Vietnamese educational culture with strong parental involvement in children's education. Language localization with primary Vietnamese interface and English support. Trust building through transparent safety features and comprehensive communication tools. The application adapts to local school structures and scheduling norms.	Which cultural characteristics could influence the approach? How do the design from different cultures differ?
Sustainability	PupilLinker is designed for long term sustainability through scalable microservices architecture that supports future growth to millions of users. The PostgreSQL database with PostGIS extensions enables horizontal scaling. Cloud based infrastructure ensures reliable operation with 93% minimum uptime. The codebase follows clean coding standards with comprehensive documentation for maintainability.	Refers to sustainability of resources, including material, energy, supplies, manufacturing techniques, personnel, operation, and the need for additional infrastructure, as well as sustainability of the design including reliability, lifetime, durability, reusability, maintainability.

14. Expected outcomes / deliverables

- PupilLinker mobile application (Beta version): A beta version for Android and iOS.
A mobile application that functions on both platforms, including the core features implemented during the development phases: multi role user management, digital attendance tracking, secure check in/check out, school bus GPS tracking, emergency alert system, parent school communication, health monitoring, incident reporting, meal management, event management, and analytics dashboard.
- Source code: The complete Frontend source code for the mobile application (Android and iOS) and the Backend source code, managed in a Git/GitHub version control repository with clear commit history and adherence to coding standards.
- Documentation:
 - + Proposal document.
 - + Project plan document.
 - + Product backlog document.
 - + User story document.
 - + Software architecture design document.
 - + Database design document.
 - + User interface design document.
 - + Meeting report document.
 - + Test plan SP1, Test plan SP2, Test plan SP3, Test plan SP4 documents.
 - + Sprint 1, Sprint 2, Sprint 3, Sprint 4 documents.
 - + Test-case_Sprint-1, Test-case_Sprint-2, Test-case_Sprint-3, Test-case_Sprint-4 documents.
 - + User guide document: Detailed instructions for end-users.
- Progress reports: Periodic reports on project progress.
- Final presentation: A presentation about the project, its features, the results achieved, and future plans.

15. References

Table 6. References

No.	References	Document Information
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		https://scrumguides.org/scrum-guide.html
2	Technical	https://docs.flutter.dev/
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		https://developer.android.com/reference/org/w3c/dom/Document
		https://developer.apple.com/documentation/xcode
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